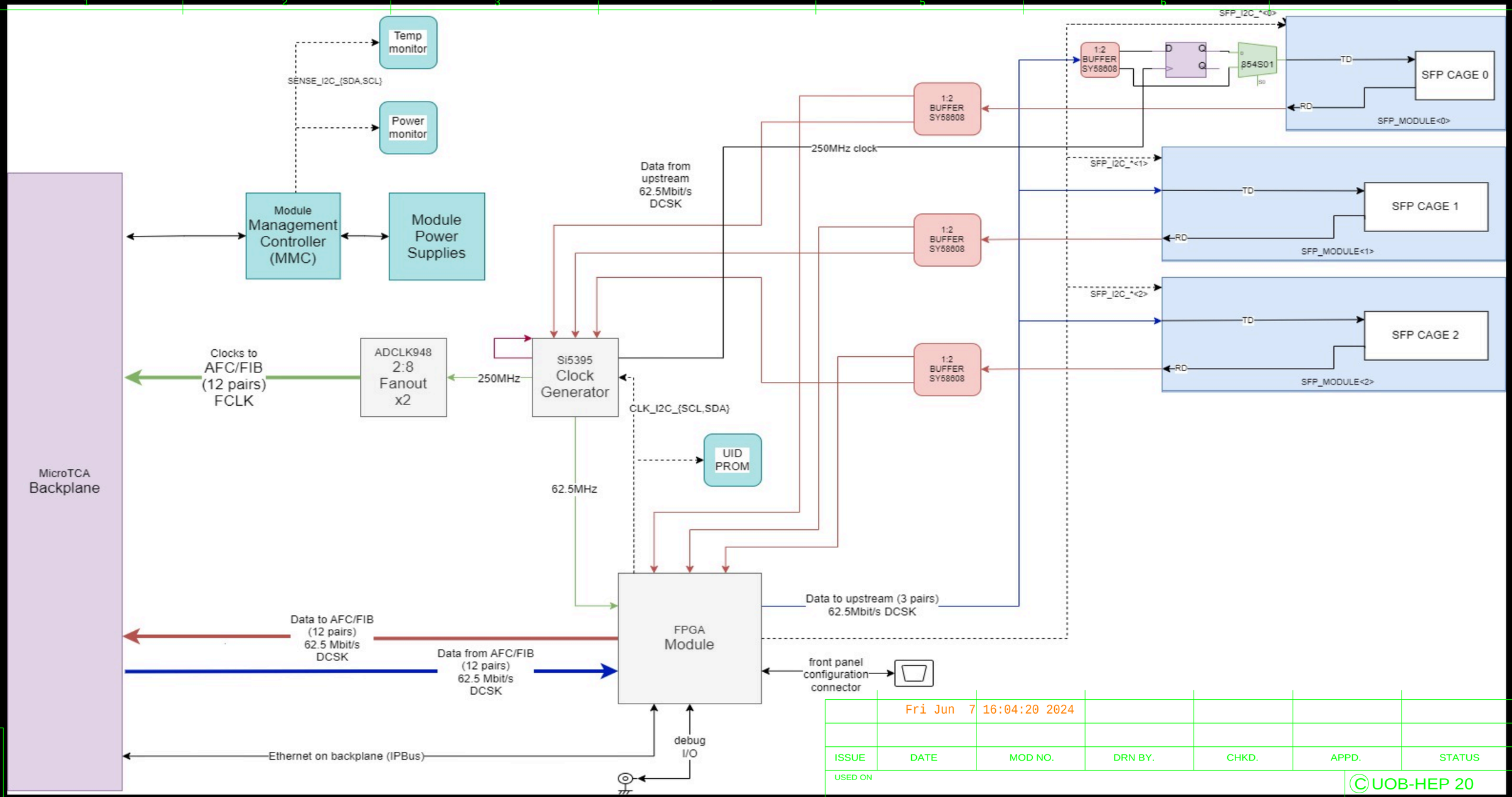


N.B. THIS IS A DRAWING, NO CONNECTIVITY

	Fri Jun 7 16:04:20 2024					
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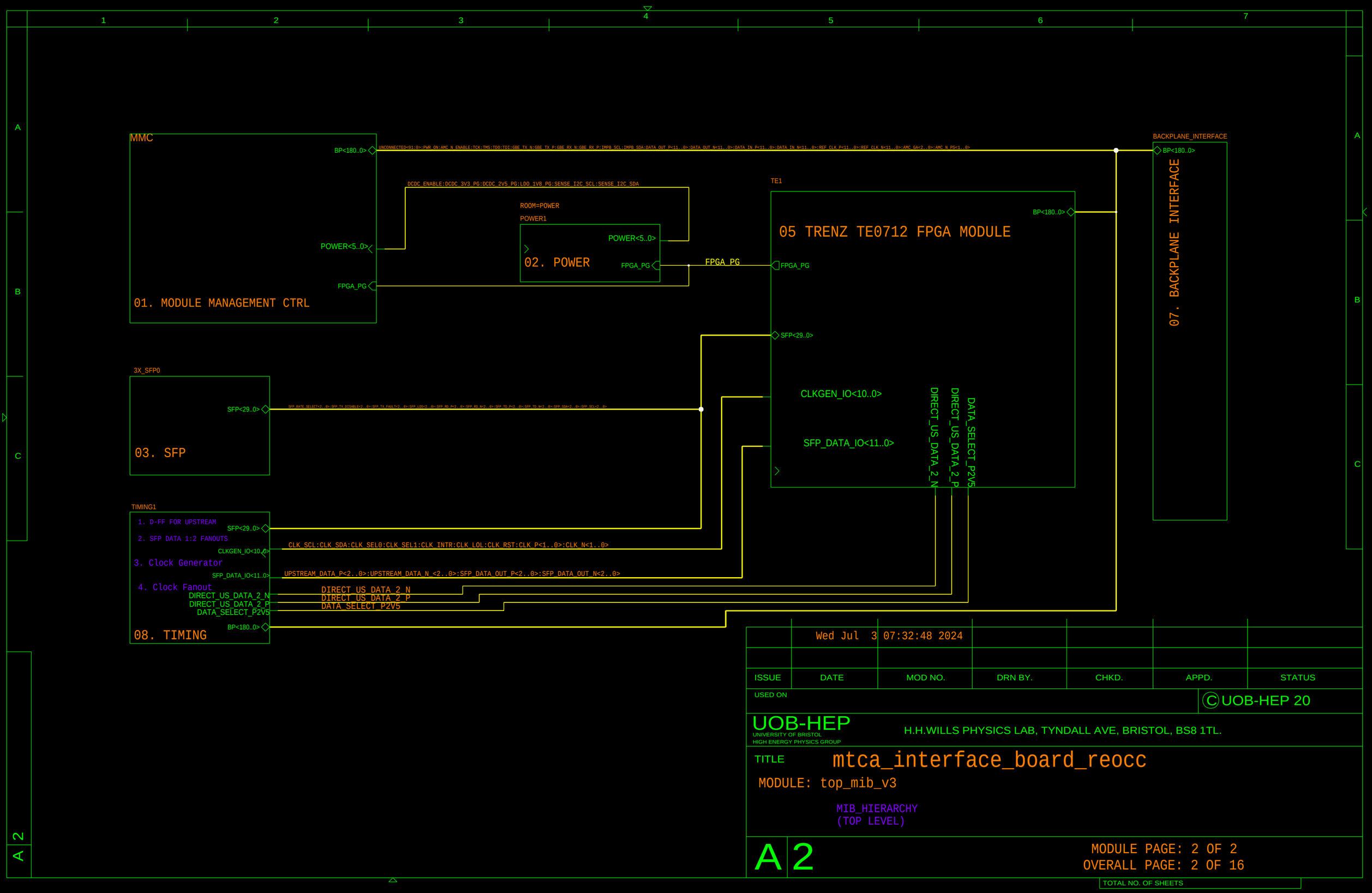
TITLE mtca_interface_board_reocc

MODULE: top_mib_v3

A2

MODULE PAGE: 1 OF 2
OVERALL PAGE: 1 OF 16

TOTAL NO. OF SHEETS



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TITLE mtca_interface_board_reocc

MODULE: top_mib_v3

MIB_HIERARCHY
(TOP LEVEL)

A2

MODULE PAGE: 2 OF 2
OVERALL PAGE: 2 OF 16

TOTAL NO. OF SHEETS

12V TO 3V3 @ 14A DC-DC CONVERTER

MIN ESR OF 3 MILLI-OHMS ON OUTPUT OF PTH08T210WAD

WANT < 0.3V DROP AT 2A

VP3V3 IV MONITOR

12V @ 6.6A IN

3.3V @ 14A OUT

VP12

C25 330UF

C51 330UF

C42 0.1UF 25V

I193

I195

POWER1

PTH08T210WAD

TRACK

TT

+SENS

VI[2]

VO[2]

UVLO

-SENS

GND[4]

VADJ

VIN<1..0>

VO_3V3<1..0>

DC_DC_ENABLE

ECP

BOM_IGNORE=TRUE

R79 XX 1%

R84 1 1%

C75 0.1UF 25V

R33 150K

T1 PJE8408 SOT523

R70 1.6K 1%

C92 0.1UF 25V

C95 100UF 6.3V

C17 220UF

C44 220UF

VCC3V3_PRE_RSHUNT

TP5

BOM_IGNORE=TRUE

R62 0.05

C61 220UF

C63 220UF

VCC3V3

IC5 TLP183 SOIC

R16 24.9

R31 24.9

MJB1

MP3V3

Default: 2-3

J12

VP12_EN

DCDC_ENABLE

HDR1x3

DCDC FORCE ENABLE

SK1

EXT POWER IN KK CONNECTOR

SENSE_12C_SDA

SENSE_12C_SCL

VCC=MP3V3

IC6 LTC2991CMS#PBF

SDA

SCL

ADR0

ADR1

ADR2

PWM

V1

V2

V3

V4

V5

V6

V7

V8

MSOP

R82

R64

R66

P2V5

VP1V8

1V8_POST_RSHUNT

P2V5_MON_HIGH

P2V5_MON_LOW

MEASURE CURRENT THROUGH
SENSE RESISTOR BEFORE 1V8 REG
MEASURE 1V8 VOLTAGE AFTER REGULATOR

I2C ADDRESS = 90H

C91 0.1UF 25V

DECOUPLE
LTC2991

R135 249 Ohm

R85 10K

R86 4.99k

D4 Red

DCDC_3V3_PG

C69 0.1UF 25V

C71 47UF 6.3V

C81 47UF 6.3V

C82 47UF 6.3V

C50 47UF 6.3V

TP7

TP13

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TITLE mtca_interface_board_reocc

MODULE: 02_ power

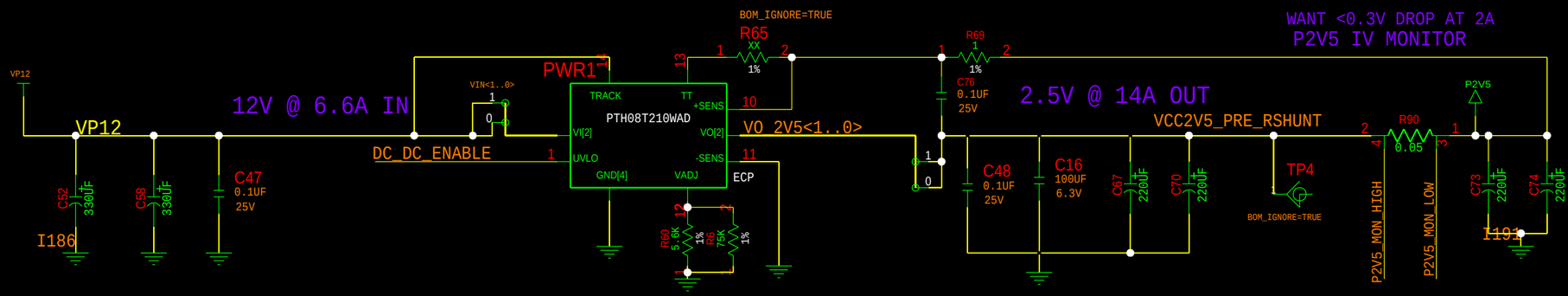
DC-DC (12V --> 3.3V)

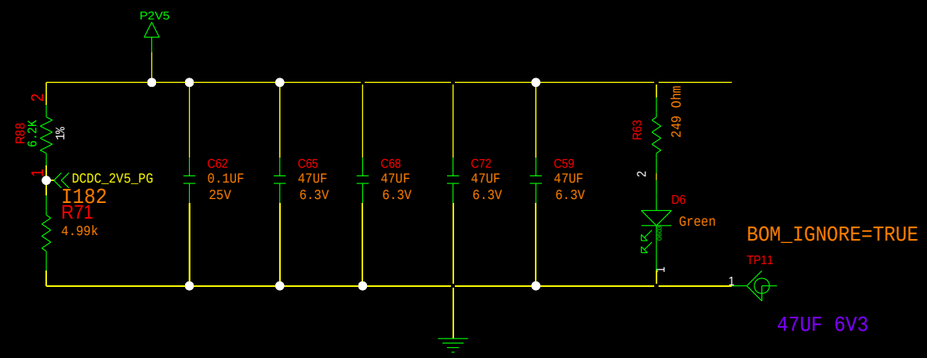
A 2

MODULE PAGE: 1 OF 3
OVERALL PAGE: 3 OF 16

TOTAL NO. OF SHEETS

12V TO 2V5 @ 14A DC-DC CONVERTER





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TITLE mtca_interface_board_reocc

MODULE: 02_ power

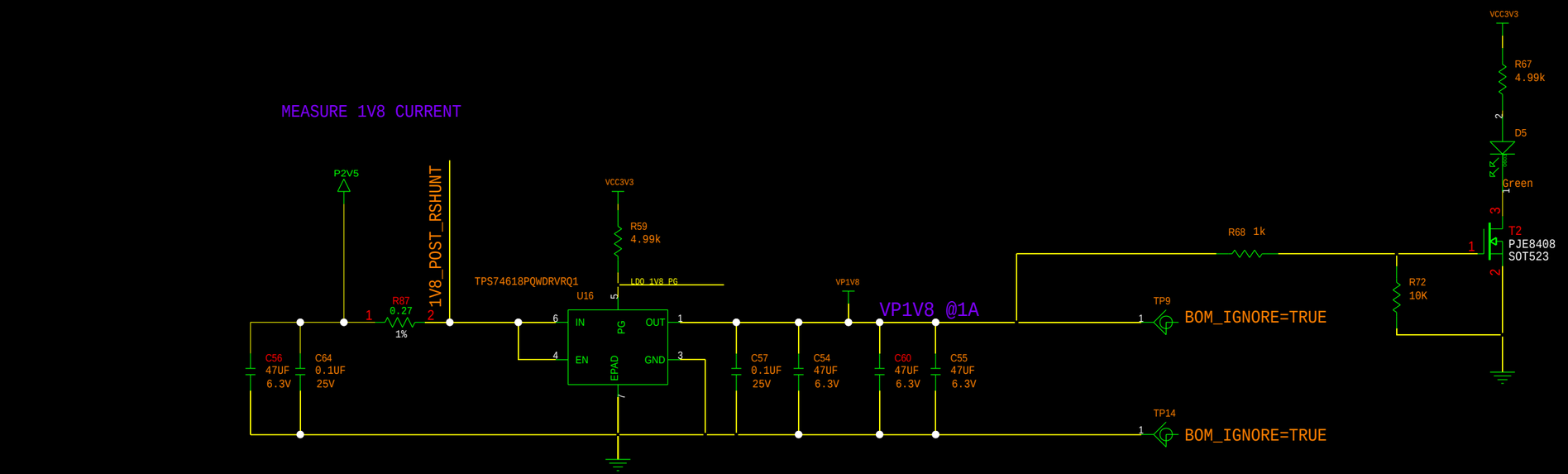
DC-DC (12V --> 2.5V)

A2

MODULE PAGE: 2 OF 3
OVERALL PAGE: 4 OF 16

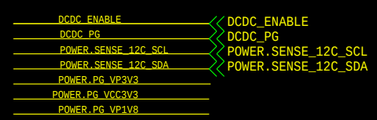
TOTAL NO. OF SHEETS

MEASURE 1V8 CURRENT



MEASURE CURRENT THROUGH
SENSE RESISTOR BEFORE 1V8 REG





FPGA_PG — FPGA_PG

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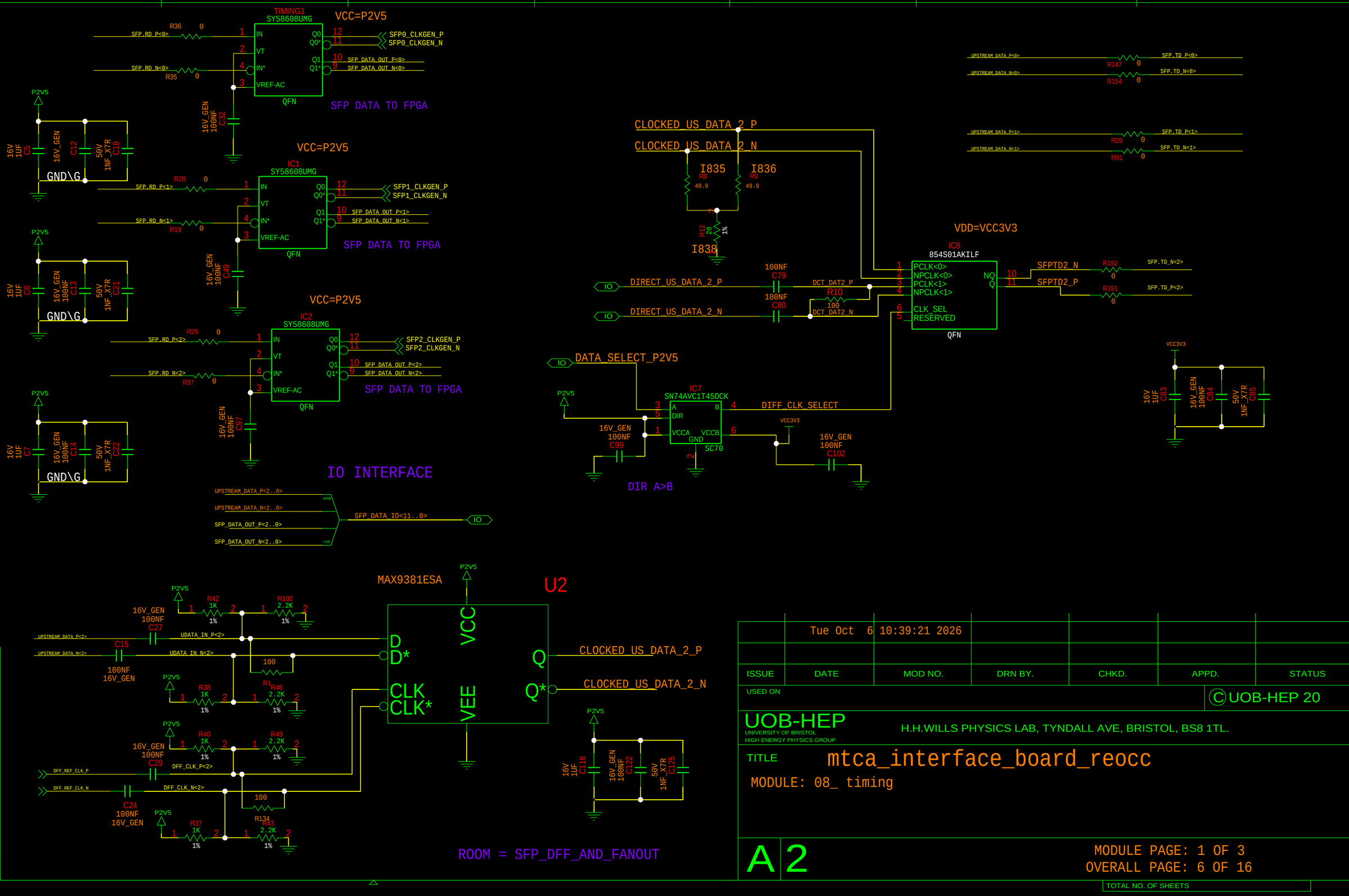
TITLE mtca_interface_board_reocc

MODULE: 02_ power

A 2

MODULE PAGE: 3 OF 3
OVERALL PAGE: 5 OF 16

TOTAL NO. OF SHEETS



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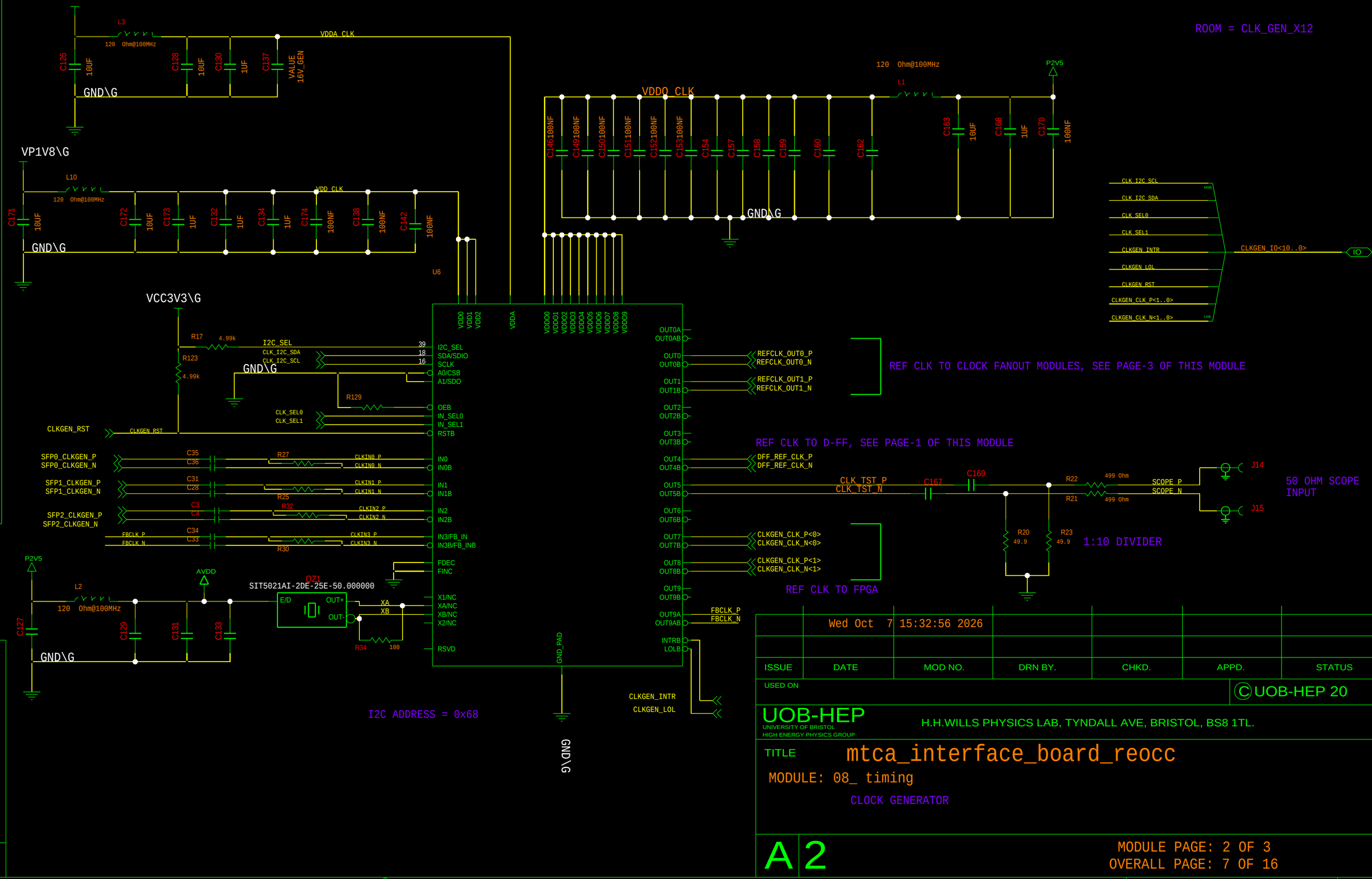
TITLE **mtca_interface_board_reocc**

MODULE: 08_ timing

A 2

MODULE PAGE: 1 OF 3
OVERALL PAGE: 6 OF 16

TOTAL NO. OF SHEETS



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TITLE mtca_interface_board_reocc

MODULE: 08_ timing

CLOCK GENERATOR

A2

VEE=GND;VCC=VDD_CLK_BUFFER

IC3 ADCLK948BCPZ

I180

C187 100NF 16V_GEN

C10 0.1UF 25V

C11 0.1UF 25V

REFCLK_OUT0_P
REFCLK_OUT0_N

RCLK_P0
RCLK_N0

VREF0 VT0 CLK0 CLK0* VT1 CLK1 CLK1* IN_SEL VREF1

LFCSP

R11 1K 1%

I213

VDD_CLK_BUFFER\G

R155 1K 1%

R157 2.2K 1%

Q[0] Q[0]* Q[1] Q[1]* Q[2] Q[2]* Q[3] Q[3]* Q[4] Q[4]* Q[5] Q[5]* Q[6] Q[6]* Q[7] Q[7]*

DC_REF_CLK_P<0>
DC_REF_CLK_N<0>
DC_REF_CLK_P<1>
DC_REF_CLK_N<1>
DC_REF_CLK_P<2>
DC_REF_CLK_N<2>
DC_REF_CLK_P<3>
DC_REF_CLK_N<3>
DC_REF_CLK_P<4>
DC_REF_CLK_N<4>
DC_REF_CLK_P<5>
DC_REF_CLK_N<5>

R161 R165 R169 R173 R177 R181 R185 R189
R159 R163 R167 R171 R175 R179 R183 R187

200 1% 2 1

C188 C189 C190 C191 C192 C193 C194 C195 C196 C197 C198 C199 100NF 16V_GEN

BP.REF_CLK_P<0>
BP.REF_CLK_N<0>
BP.REF_CLK_P<1>
BP.REF_CLK_N<1>
BP.REF_CLK_P<2>
BP.REF_CLK_N<2>
BP.REF_CLK_P<3>
BP.REF_CLK_N<3>
BP.REF_CLK_P<4>
BP.REF_CLK_N<4>
BP.REF_CLK_P<5>
BP.REF_CLK_N<5>

ROOM = CLK_FO_0_5

VEE=GND;VCC=VDD_CLK_BUFFER

IC4 ADCLK948BCPZ

I247

C139 100NF 16V_GEN

C26 0.1UF 25V

C30 0.1UF 25V

REFCLK_OUT1_P
REFCLK_OUT1_N

RCLK_P1
RCLK_N1

VREF0 VT0 CLK0 CLK0* VT1 CLK1 CLK1* IN_SEL VREF1

LFCSP

R24 1K 1%

I254

VDD_CLK_BUFFER\G

R156 1K 1%

R158 2.2K 1%

Q[0] Q[0]* Q[1] Q[1]* Q[2] Q[2]* Q[3] Q[3]* Q[4] Q[4]* Q[5] Q[5]* Q[6] Q[6]* Q[7] Q[7]*

DC_REF_CLK_P<6>
DC_REF_CLK_N<6>
DC_REF_CLK_P<7>
DC_REF_CLK_N<7>
DC_REF_CLK_P<8>
DC_REF_CLK_N<8>
DC_REF_CLK_P<9>
DC_REF_CLK_N<9>
DC_REF_CLK_P<10>
DC_REF_CLK_N<10>
DC_REF_CLK_P<11>
DC_REF_CLK_N<11>

R162 R166 R170 R174 R178 R182 R186 R190
R160 R164 R168 R172 R176 R180 R184 R188

200 1% 2 1

I243

C175 C176 C177 C178 C179 C180 C181 C182 C183 C184 C185 C186 100NF 16V_GEN

BP.REF_CLK_P<6>
BP.REF_CLK_N<6>
BP.REF_CLK_P<7>
BP.REF_CLK_N<7>
BP.REF_CLK_P<8>
BP.REF_CLK_N<8>
BP.REF_CLK_P<9>
BP.REF_CLK_N<9>
BP.REF_CLK_P<10>
BP.REF_CLK_N<10>
BP.REF_CLK_P<11>
BP.REF_CLK_N<11>

ROOM = CLK_FO_6_11

I214 VCC3V3

L9 BLM21PG221SN1

VDD_CLK_BUFFER\G

GND\G

C112 10UF 10V

C121 10UF 10V C37 100NF 16V_GEN

C38 1UF 6.3V C135 100NF 16V_GEN C19 100NF 16V_GEN

C20 1UF 6.3V C136 100NF 16V_GEN

C155 10UF 10V C148 100NF 16V_GEN

C165 1UF 6.3V C166 100NF 16V_GEN C167 100NF 16V_GEN

C9 1UF 6.3V C8 100NF 16V_GEN

I177 SFP<29..0>

SFP<29..0>

SFP_RATE_SELECT<2..0>
SFP_TX_DISABLE<2..0>
SFP_TX_FAULT<2..0>
SFP_LOS<2..0>
SFP_RD_P<2..0>
SFP_RD_N<2..0>
SFP_TD_P<2..0>
SFP_TD_N<2..0>
SFP_SDA<2..0>
SFP_SCL<2..0>

BP<180..0>

BP.REF_CLK_P<11..0>
BP.REF_CLK_N<11..0>

USE 8 WAY LVPECL FANOUT
... BECAUSE 6 WAY LVDS BUFFER
NOT AVAILABLE DUE TO
SUPPLY CHAIN ISSUES

ADCLK984 SUPPLY CURRENT < 330MA
10MA PER OUTPUT PIN = 160MA
TOTAL PER DEVICE = 490MA
TOTAL FOR FANOUT = 980MA

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TITLE mtca_interface_board_reocc

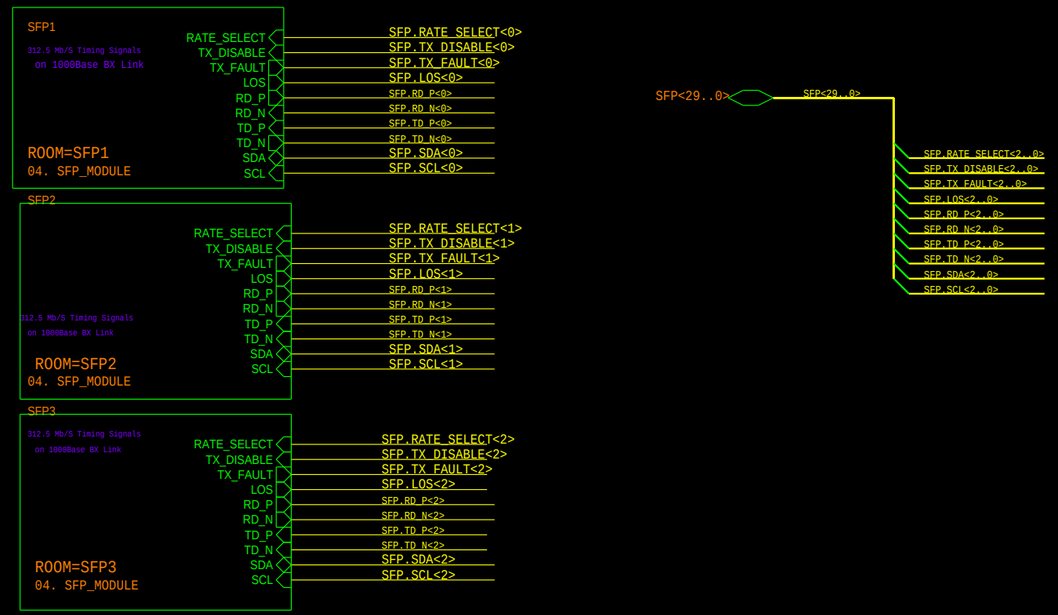
MODULE: 08_ timing

CLOCK FANOUT

A2

MODULE PAGE: 3 OF 3
OVERALL PAGE: 8 OF 16

TOTAL NO. OF SHEETS



	Mon Mar 6 16:31:19 2023					
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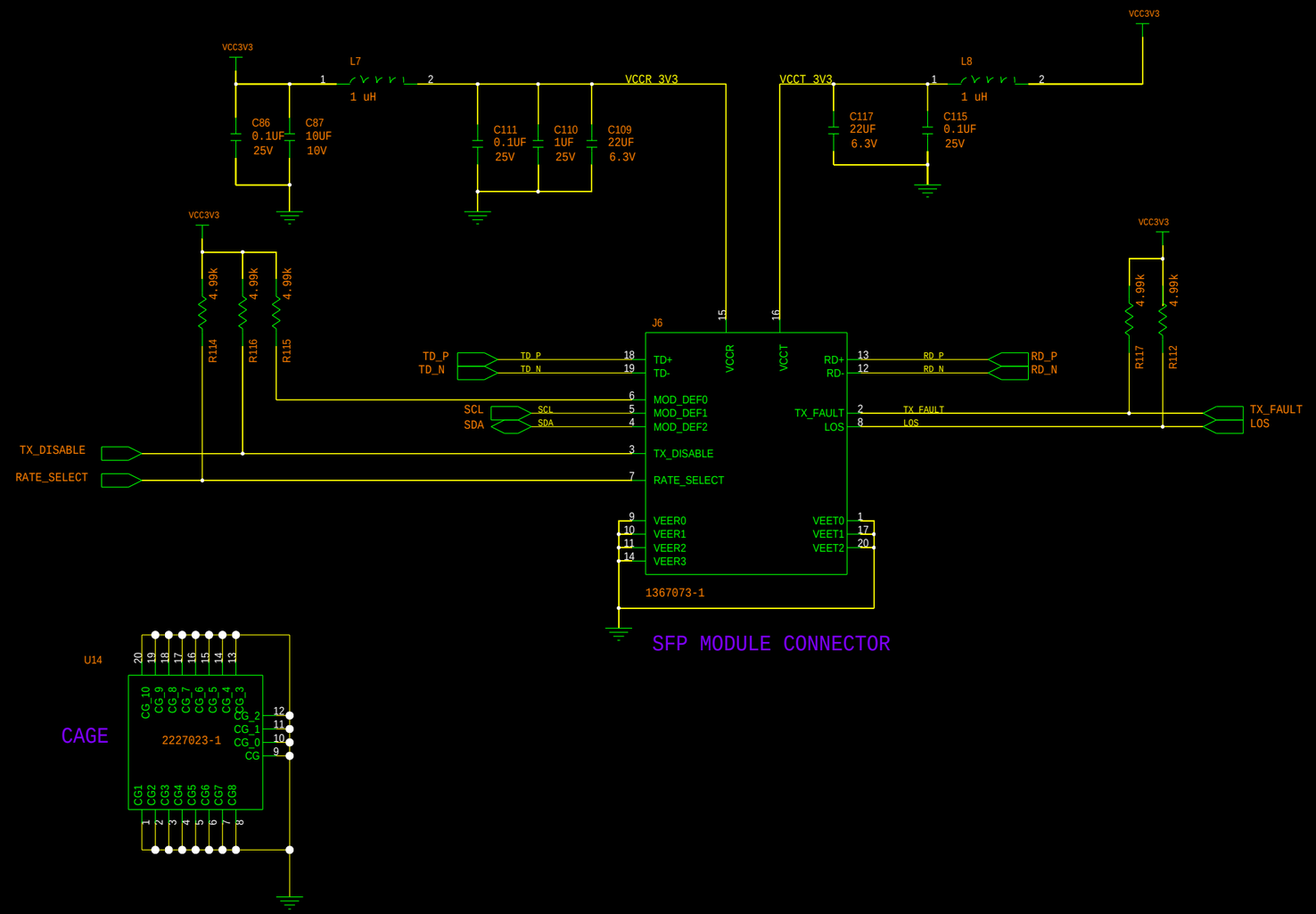
TITLE mtca_interface_board_reocc

MODULE: 03_ sfp

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 9 OF 16

TOTAL NO. OF SHEETS



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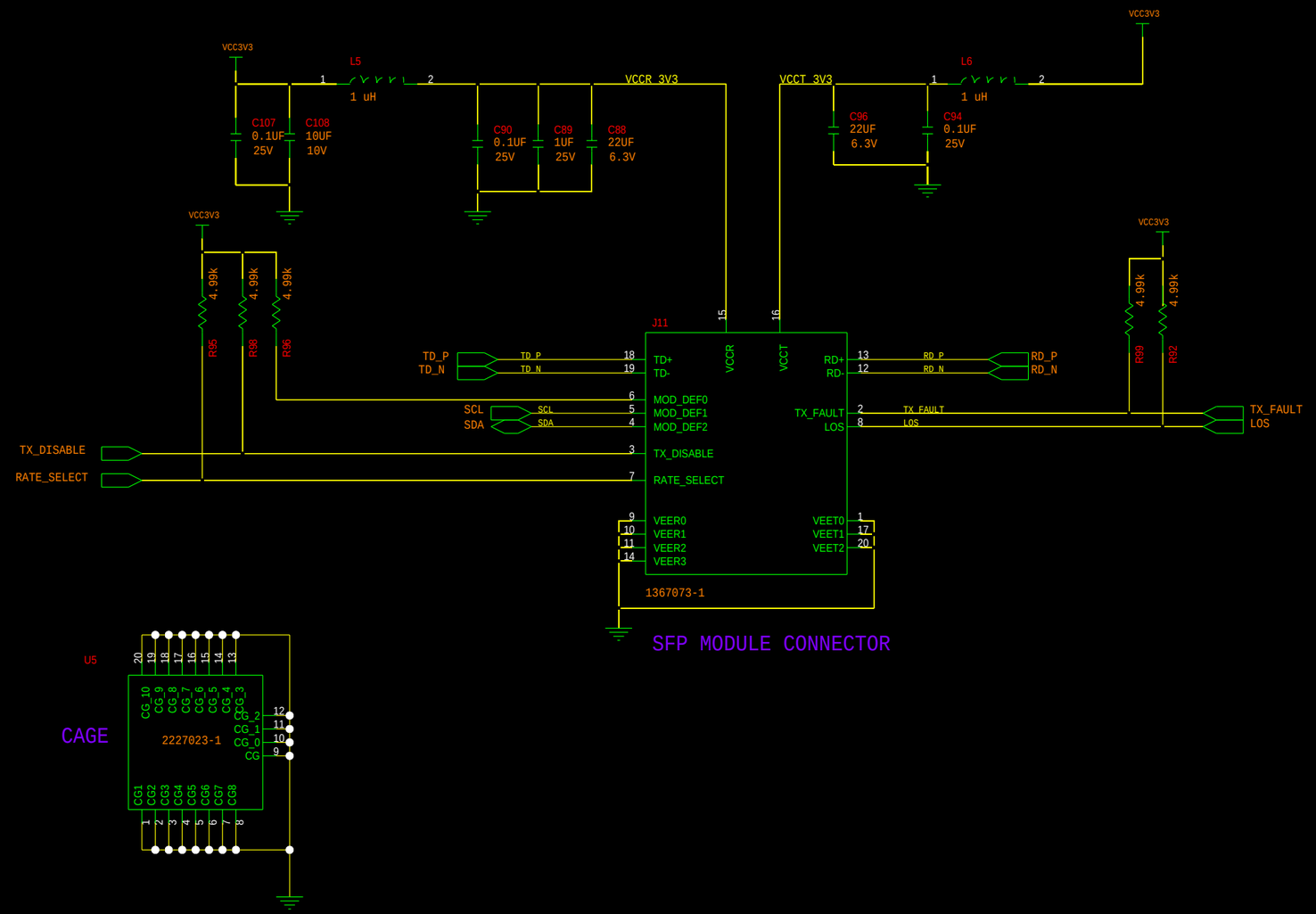
TITLE mtca_interface_board_reocc

MODULE: 04_ sfp_module

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 10 OF 16

TOTAL NO. OF SHEETS



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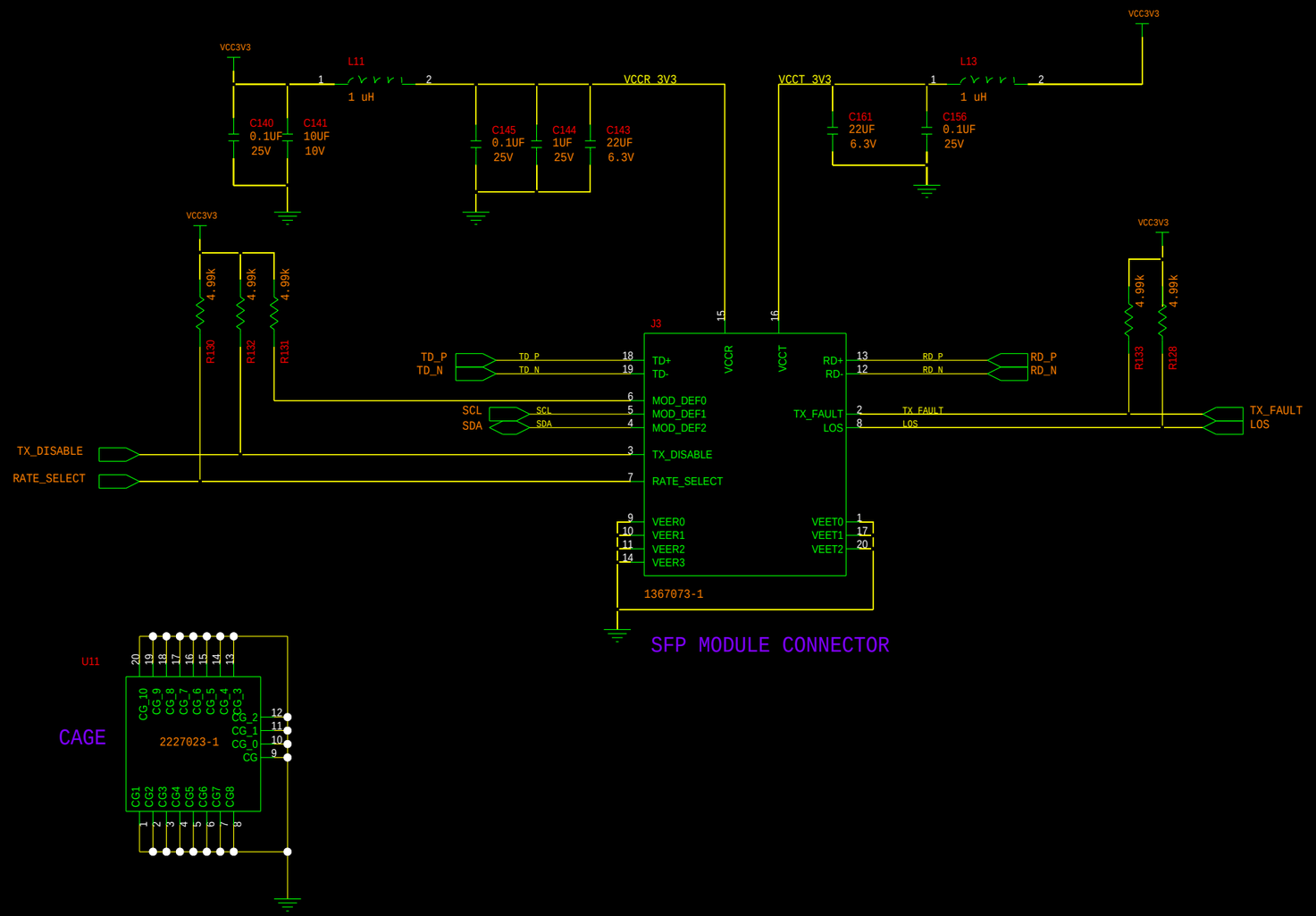
TITLE mtca_interface_board_reocc

MODULE: 04_ sfp_module

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 11 OF 16

TOTAL NO. OF SHEETS



SFP MODULE CONNECTOR

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TITLE mtca_interface_board_reocc

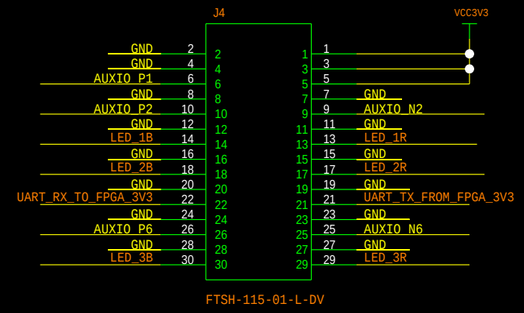
MODULE: 04_ sfp_module

A2

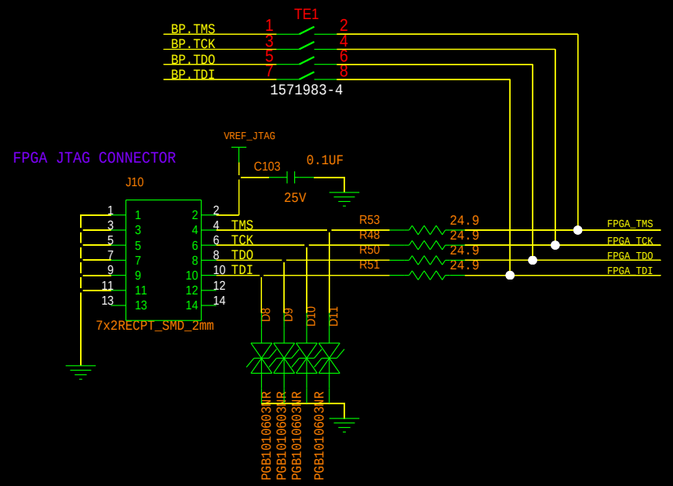
MODULE PAGE: 1 OF 1
OVERALL PAGE: 12 OF 16

TOTAL NO. OF SHEETS

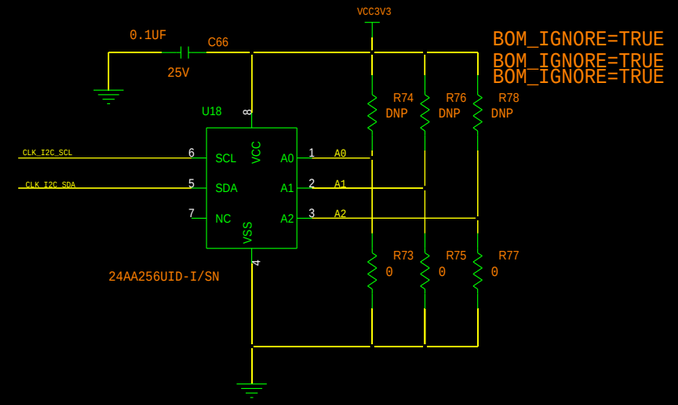
AUX IO INTERFACE



FPGA JTAG CONNECTOR

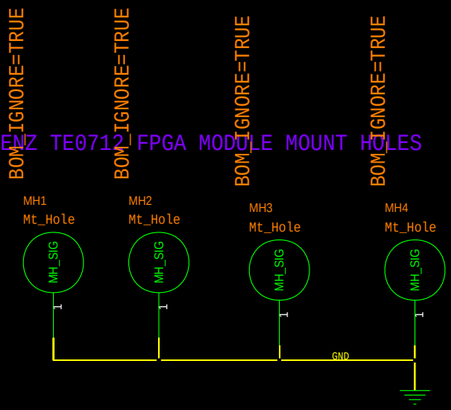


EEPROM

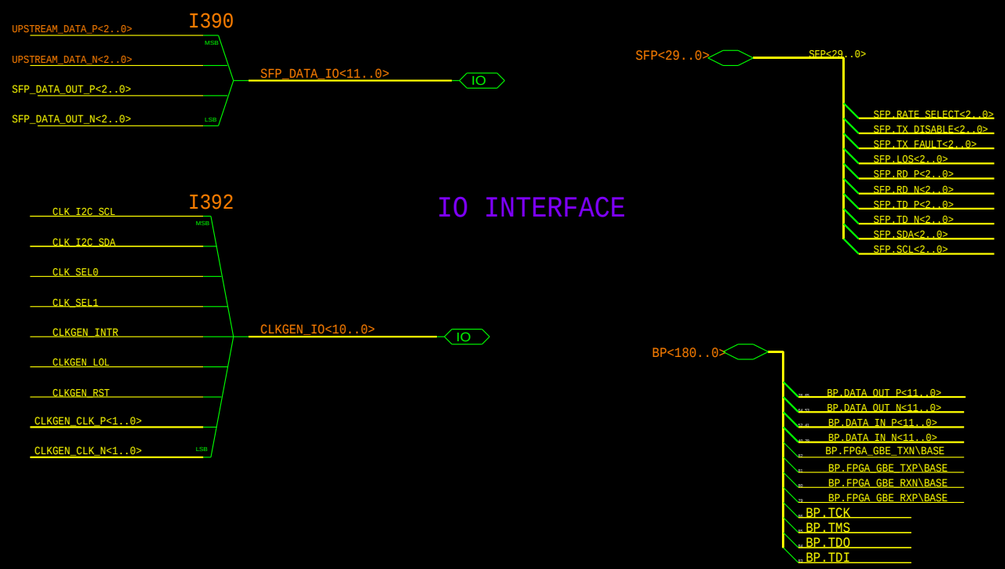


I2C ADDRESS = 0X50

TRENZ TE0712 FPGA MODULE MOUNT HOLES



IO INTERFACE



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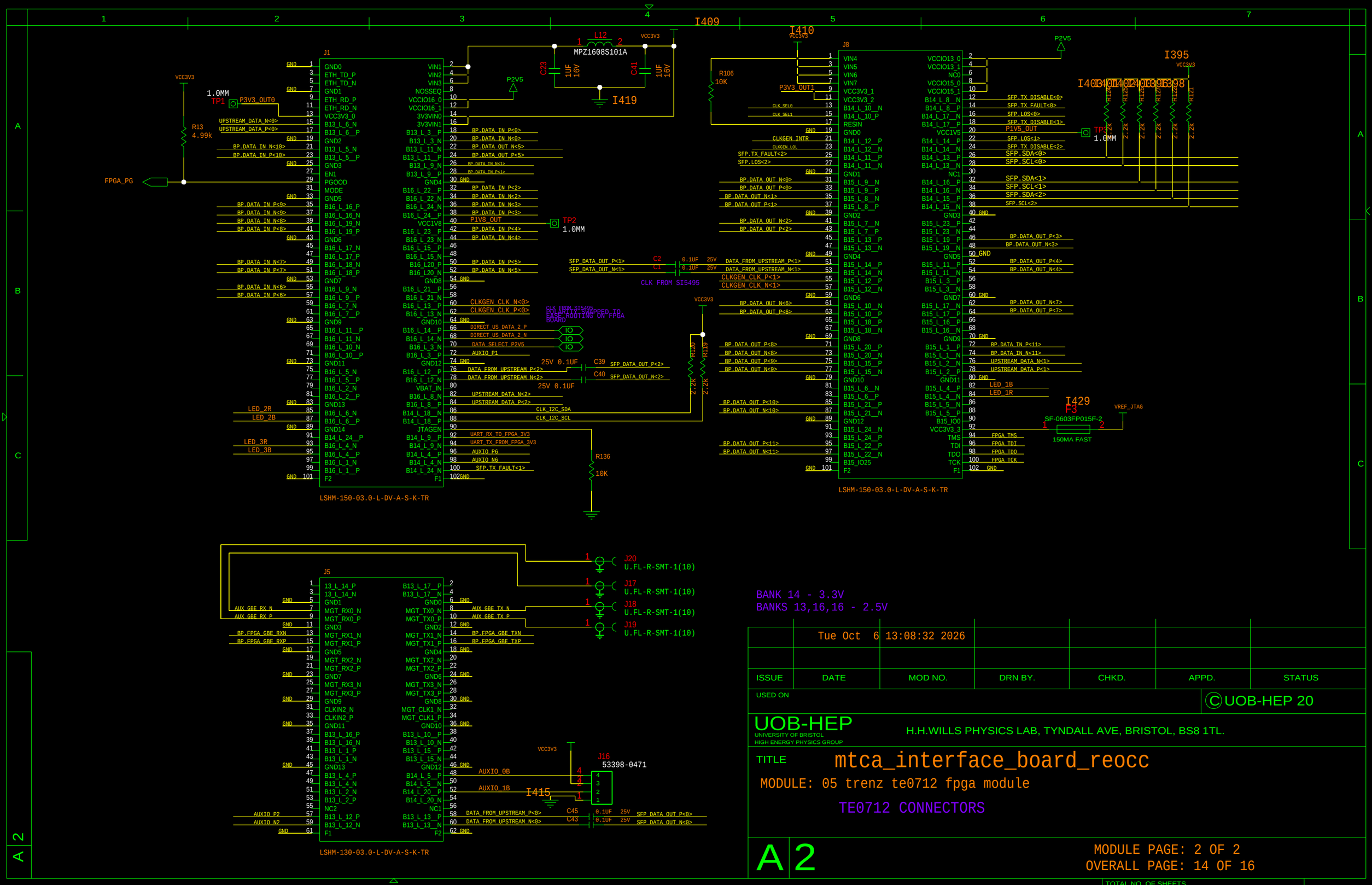
TITLE **mtca_interface_board_reocc**

MODULE: 05 trenz te0712 fpga module

A2

MODULE PAGE: 1 OF 2
OVERALL PAGE: 13 OF 16

TOTAL NO. OF SHEETS



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TITLE **mtca_interface_board_reocc**

MODULE: 05 trenz te0712 fpga module

TE0712 CONNECTORS

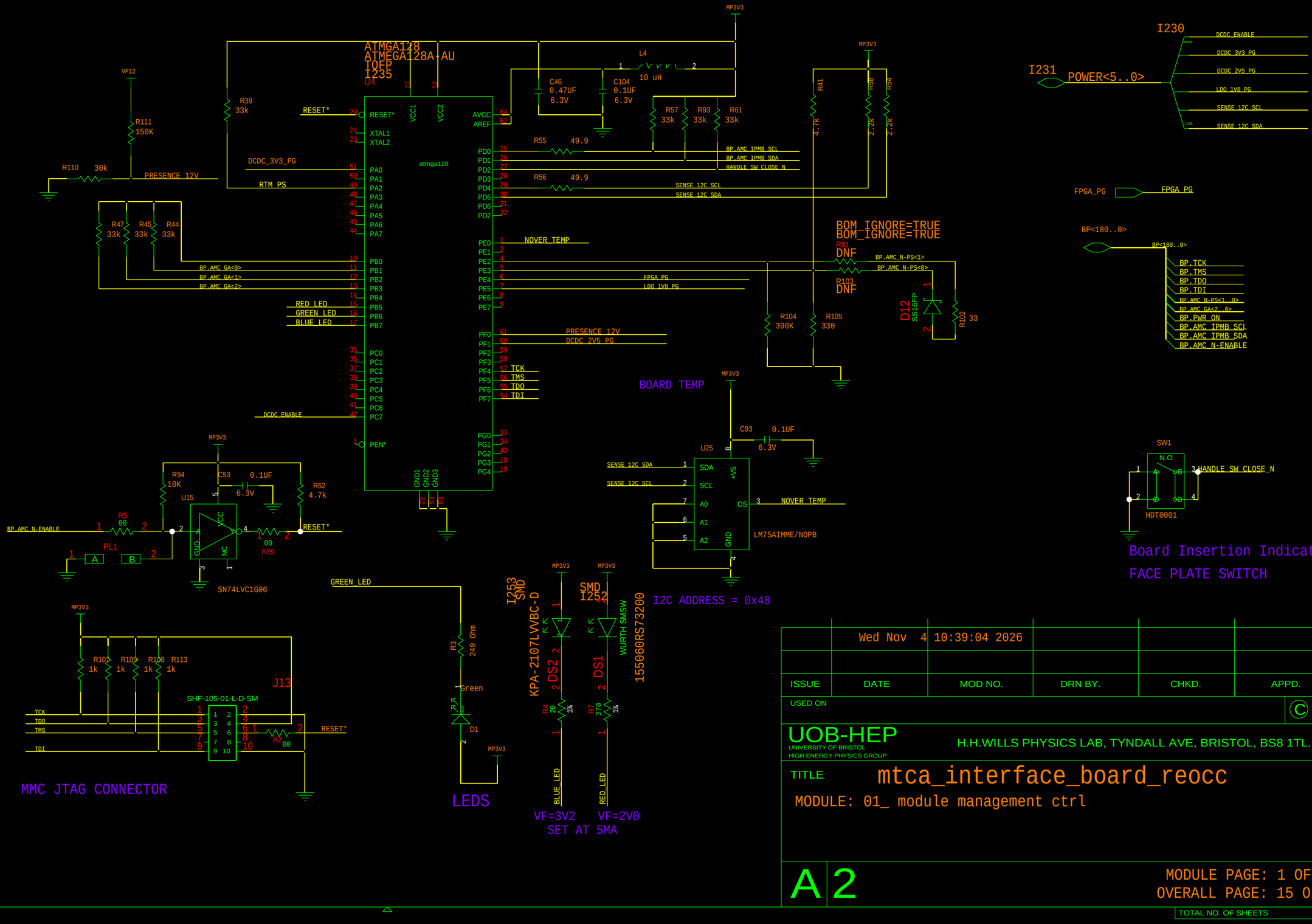
A 2

MODULE PAGE: 2 OF 2
OVERALL PAGE: 14 OF 16

TOTAL NO. OF SHEETS

MODULE MANAGEMENT CONTROL

IO INTERFACE



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TITLE **mtca_interface_board_reocc**

MODULE: 01_ module management ctrl

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 15 OF 16

TOTAL NO. OF SHEETS

TONGUE 2 MEZZENINE CONNECTOR

VP12
F1
SF-0603S300-2
3A
MP3V3
F2
SF-0603FP015F-2
150MA FAST

BOM_IGNORE=TRUE

J7

GND	1	GND1	PWR_ON	170	BP.PWR_ON
12V_IN	2	PWR1	TDI	169	BP.TDI
		PS1#	TDO	168	BP.TDO
MP3V3_IN	4	MP	TRST#	167	
BP.AMC_GA<0>	5	GA0	TMS	166	BP.TMS
	6	RSVD0	TCK	165	BP.TCK
GND	7	GND2	GND55	164	GND
	8	RSVD1	TXFA-1+	163	
12V_IN	9	PWR2	TXFA-1-	162	
GND	10	GND3	GND54	161	GND
BP.FPGA_GBE_TXP	11	TXFUA+	RXFA-1+	160	
BP.FPGA_GBE_TXN	12	TXFUA-	RXFA-1-	159	
GND	13	GND4	GND53	158	GND
BP.FPGA_GBE_RXP	14	RXFUA+	TXFA-2+	157	
BP.FPGA_GBE_RXN	15	RXFUA-	TXFA-2-	156	
GND	16	GND5	GND52	155	GND
BP.AMC_GA<1>	17	GA1	RXFA-2+	154	
12V_IN	18	PWR3	RXFA-2-	153	
GND	19	GND6	GND51	152	GND
	20	TXFA-3+	TXFA-4+	151	
	21	TXFA-3-	TXFA-4-	150	
GND	22	GND7	GND50	149	GND
	23	RXFA-3+	RXFA-4+	148	
	24	RXFA-3-	RXFA-4-	147	
GND	25	GND8	GND49	146	GND
BP.AMC_GA<2>	26	GA2	TXFA-6+	145	
12V_IN	27	PWR4	TXFA-6-	144	
GND	28	GND9	GND48	143	GND
	29	TXFA-5+	RXFA-6+	142	
	30	TXFA-5-	RXFA-6-	141	
GND	31	GND10	GND47	140	GND
	32	RXFA-5+	TXFA-8+	139	
	33	RXFA-5-	TXFA-8-	138	
GND	34	GND11	GND46	137	GND
	35	TXFA-7+	RXFA-8+	136	
	36	TXFA-7-	RXFA-8-	135	
GND	37	GND12	GND45	134	GND
	38	RXFA-7+	TMREQ#	133	
	39	RXFA-7-	RSVD	132	
GND	40	GND13	GND44	131	GND
BP.AMC_N-ENABLE	41	ENABLE#	I2C_SCL	130	
12V_IN	42	PWR5	I2C_SDA	129	
GND	43	GND14	GND43	128	GND
	44	TXFA-9+	IPMB0-SCL-A	127	
	45	TXFA-9-	IPMB0-SDA-A	126	
GND	46	GND15	GND42	125	GND
	47	RXFA-9+	IPMB0-SCL-B	124	
	48	RXFA-9-	IPMB0-SDA-B	123	
GND	49	GND16	GND41	122	GND
	50	TXFA-10+	IPMBL-SCL-1	121	
	51	TXFA-10-	IPMBL-SDA-1	120	
GND	52	GND17	GND40	119	GND
	53	RXFA-10+	IPMBL-SCL-2	118	
	54	RXFA-10-	IPMBL-SDA-2	117	
GND	55	GND18	GND39	116	GND
BP.AMC_IPMB_SCL	56	SCL_L	IPMBL-SCL-3	115	
12V_IN	57	PWR6	IPMBL-SDA-3	114	
GND	58	GND19	GND38	113	GND
	59	TXFA-11+	IPMBL-SCL-4	112	
	60	TXFA-11-	IPMBL-SDA-4	111	
GND	61	GND20	GND37	110	GND
	62	RXFA-11+	IPMBL-SCL-5	109	
	63	RXFA-11-	IPMBL-SDA-5	108	
GND	64	GND21	GND36	107	GND
	65	TXFA-12+	IPMBL-SCL-6	106	
	66	TXFA-12-	IPMBL-SDA-6	105	
GND	67	GND22	GND35	104	GND
	68	RXFA-12+	IPMBL-SCL-7	103	
	69	RXFA-12-	IPMBL-SDA-7	102	
GND	70	GND23	GND34	101	GND
BP.AMC_IPMB_SDA	71	SDA_L	IPMBL-SCL-8	100	
12V_IN	72	PWR7	IPMBL-SDA-8	99	
GND	73	GND24	GND33	98	GND
	74	XOVER0+	IPMBL-SCL-9	97	
	75	XOVER0-	IPMBL-SDA-9	96	
GND	76	GND25	GND32	95	GND
	77	XOVER1+	IPMBL-SCL-10	94	
	78	XOVER1-	IPMBL-SDA-10	93	
GND	79	GND26	GND31	92	GND
	80	XOVER2+	IPMBL-SCL-11	91	
	81	XOVER2-	IPMBL-SDA-11	90	
GND	82	GND27	GND30	89	GND
BP.AMC_N-PS<0>	83	PS0#	IPMBL-SCL-12	88	
12V_IN	84	PWR8	IPMBL-SDA-12	87	
GND	85	GND28	GND29	86	GND

MCH_TG1-EDGE-170

BP PLANE CONNECTOR TONGUE 1

R80
0

J2

2	1	1	GND	2	GND
4	3	3	GND	4	BP.REF_CLK_P<0>
6	5	5	GND	6	BP.REF_CLK_N<0>
8	7	7	BP.REF_CLK_P<1>	8	GND
10	9	9	BP.REF_CLK_N<1>	10	GND
12	11	11	GND	12	BP.REF_CLK_P<2>
14	13	13	GND	14	BP.REF_CLK_N<2>
16	15	15	BP.REF_CLK_P<3>	16	GND
18	17	17	BP.REF_CLK_N<3>	18	GND
20	19	19	GND	20	BP.REF_CLK_P<4>
22	21	21	GND	22	BP.REF_CLK_N<4>
24	23	23	BP.REF_CLK_P<5>	24	GND
26	25	25	BP.REF_CLK_N<5>	26	GND
28	27	27	GND	28	BP.REF_CLK_P<6>
30	29	29	GND	30	BP.REF_CLK_N<6>
32	31	31	BP.REF_CLK_P<7>	32	GND
34	33	33	BP.REF_CLK_N<7>	34	GND
36	35	35	GND	36	BP.REF_CLK_P<8>
38	37	37	GND	38	BP.REF_CLK_N<8>
40	39	39	BP.REF_CLK_P<9>	40	GND
42	41	41	BP.REF_CLK_N<9>	42	GND
44	43	43	GND	44	BP.REF_CLK_P<10>
46	45	45	GND	46	BP.REF_CLK_N<10>
48	47	47	BP.REF_CLK_P<11>	48	GND
50	49	49	BP.REF_CLK_N<11>	50	GND
52	51	51	GND	52	GND
54	53	53	GND	54	GND
56	55	55	GND	56	GND
58	57	57	GND	58	GND
60	59	59	GND	60	GND
F2	F1	61	GND	62	GND

LSHM-130-03.0-L-DV-A-S-K-TR

BP<180..0>

IO INTERFACE

BP.PWR_ON
BP.AMC_N-ENABLE
BP.TCK
BP.TMS
BP.TDO
BP.TDI
BP.FPGA_GBE_TXN
BP.FPGA_GBE_TXP
BP.FPGA_GBE_RXN
BP.FPGA_GBE_RXP
BP.AMC_IPMB_SCL
BP.AMC_IPMB_SDA
BP.DATA_OUT_P<11..0>
BP.DATA_OUT_N<11..0>
BP.DATA_IN_P<11..0>
BP.DATA_IN_N<11..0>
BP.REF_CLK_P<11..0>
BP.REF_CLK_N<11..0>
BP.AMC_GA<2..0>
BP.AMC_N-PS<1..0>

BOM_IGNORE=TRUE
BOM_IGNORE=TRUE

ESD PROTECTION

SEGMENT2
SEGMENT1
SEGMENT3
R138
10M
R137
10M

Promotes save ESD discharge
during board insertion (see pg

2-16) of PICMG_ATCA

FRONT PANEL MOUNT HOLES

MH10
Mt_Hole
MH_SIG
MH5
Mt_Hole
MH_SIG

BOM_IGNORE=TRUE
BOM_IGNORE=TRUE
BOM_IGNORE=TRUE
BOM_IGNORE=TRUE

J9

GND	2	2	1	1	GND
BP.DATA_IN_P<0>	4	4	3	3	GND
BP.DATA_IN_N<0>	6	6	5	5	GND
GND	8	8	7	7	BP.DATA_IN_P<1>
GND	10	10	9	9	BP.DATA_IN_N<1>
BP.DATA_IN_P<2>	12	12	11	11	GND
BP.DATA_IN_N<2>	14	14	13	13	GND
GND	16	16	15	15	BP.DATA_IN_P<3>
GND	18	18	17	17	BP.DATA_IN_N<3>
BP.DATA_IN_P<4>	20	20	19	19	GND
BP.DATA_IN_N<4>	22	22	21	21	GND
GND	24	24	23	23	BP.DATA_IN_P<5>
GND	26	26	25	25	BP.DATA_IN_N<5>
BP.DATA_IN_P<6>	28	28	27	27	GND
BP.DATA_IN_N<6>	30	30	29	29	GND
GND	32	32	31	31	BP.DATA_IN_P<7>
GND	34	34	33	33	BP.DATA_IN_N<7>
BP.DATA_IN_P<8>	36	36	35	35	GND
BP.DATA_IN_N<8>	38	38	37	37	GND
GND	40	40	39	39	BP.DATA_IN_P<9>
GND	42	42	41	41	BP.DATA_IN_N<9>
BP.DATA_IN_P<10>	44	44	43	43	GND
BP.DATA_IN_N<10>	46	46	45	45	GND
GND	48	48	47	47	BP.DATA_IN_P<11>
GND	50	50	49	49	BP.DATA_IN_N<11>
BP.DATA_OUT_P<0>	52	52	51	51	GND
BP.DATA_OUT_N<0>	54	54	53	53	GND
GND	56	56	55	55	BP.DATA_OUT_P<1>
GND	58	58	57	57	BP.DATA_OUT_N<1>
BP.DATA_OUT_P<2>	60	60	59	59	GND
BP.DATA_OUT_N<2>	62	62	61	61	GND
GND	64	64	63	63	BP.DATA_OUT_P<3>
GND	66	66	65	65	BP.DATA_OUT_N<3>
BP.DATA_OUT_P<4>	68	68	67	67	GND
BP.DATA_OUT_N<4>	70	70	69	69	GND
GND	72	72	71	71	BP.DATA_OUT_P<5>
GND	74	74	73	73	BP.DATA_OUT_N<5>
BP.DATA_OUT_P<6>	76	76	75	75	GND
BP.DATA_OUT_N<6>	78	78	77	77	GND
GND	80	80	79	79	BP.DATA_OUT_P<7>
GND	82	82	81	81	BP.DATA_OUT_N<7>
BP.DATA_OUT_P<8>	84	84	83	83	GND
BP.DATA_OUT_N<8>	86	86	85	85	GND
GND	88	88	87	87	BP.DATA_OUT_P<9>
GND	90	90	89	89	BP.DATA_OUT_N<9>
BP.DATA_OUT_P<10>	92	92	91	91	GND
BP.DATA_OUT_N<10>	94	94	93	93	GND
GND	96	96	95	95	BP.DATA_OUT_P<11>
GND	98	98	97	97	BP.DATA_OUT_N<11>
GND	100	100	99	99	GND
F2			F1	101	GND

LSHM-150-03.0-L-DV-A-S-K-TR

MH8 Mt_Hole MH_SIG
MH9 Mt_Hole MH_SIG
MH6 Mt_Hole MH_SIG
MH7 Mt_Hole MH_SIG

TONGUE 2 MEZZENINE MOUNT HOLES

	Wed Nov 4 10:39:04 2026					
ISSUE	DATE	MOD NO.	DRN BY.	CHKD.	APPD.	STATUS

USED ON

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UNIVERSITY OF BRISTOL
HIGH ENERGY PHYSICS GROUP
H.H.WILLS PHYSICS LAB, TYNDALL AVE, BRISTOL, BS8 1TL.

TITLE mtca_interface_board_reocc

MODULE: 07_ backplane interface

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 16 OF 16

TOTAL NO. OF SHEETS